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Abstract

Background: Air pollution and the rising prevalence of liver diseases have become public health concerns. Nevertheless, the relationship and underlying mechanisms between ambient air pollution and liver function remain inadequately investigated.

Methods: A total of 4,096 rural residents living in the suburb of Xinxiang city, China were recruited from the Henan Rural Cohort. The participants underwent two physical examinations conducted in 2017 and 2021, respectively. A generalized linear model was utilized to evaluate the associations between individual ambient air pollutants and liver function biomarkers. To determine the main air pollutants that have impacts on liver function biomarkers, weighted quantile sum (WQS) regression models were applied. Additionally, the mediating role of fasting insulin (FINS) in these associations was investigated using mediation analysis.

Results: For each increase in the interquartile range (IQR) of fine particulate matter (PM_{2.5}) (lag03, representing the average exposure over the current and previous 3 days) concentration (IQR: 17.73 $\mu\text{g}/\text{m}^3$), aspartate aminotransferase levels increased by 2.30% (95% confidence interval [CI]: 1.34%, 3.28%). Similar relationships were also detected between other air pollutants and biomarkers of liver function. WQS regression analysis confirmed that sulfur dioxide (weight: 0.64), nitrogen dioxide (NO₂) (0.13), and ozone (0.18) were the main contributing pollutants associated with total bilirubin (TBIL). FINS was significantly associated with NO₂ exposure and increases in indirect bilirubin (IBIL) (11.41%, 95% CI: 9.78%, 16.58%), direct bilirubin (DBIL) (5.32%, 95% CI: 5.02%, 8.47%), and TBIL (11.56%, 95% CI: 9.97%, 16.58%). The mediating effect of FINS in the relationship between PM_{2.5} and alanine aminotransferase levels were significant. It accounted for -49.92% (95% CI: -96.56%, -29.99%) of the total effect.

Conclusions: There is a positive association between short-term exposures to ambient air pollutants and biomarkers of liver function. Moreover, FINS may play mediating roles in these associations.

Keywords: Air pollution; Liver function; Fasting insulin; Mediation

Background

Approximately 2 million individuals worldwide die from liver diseases annually, imposing a significant economic and social burden on the global population (1). Accumulating evidence indicates that air pollutants may contribute to liver dysfunction through systemic oxidative stress and inflammatory mechanisms (2, 3). Specifically, inhaled pollutants such as fine particulate matter (PM_{2.5}, with an aerodynamic diameter ≤ 2.5 μm), nitrogen dioxide (NO₂), and ozone (O₃) can induce the release of pro-inflammatory cytokines and reactive oxygen species into the bloodstream, potentially leading to direct hepatocyte damage and the promotion of hepatic steatosis (4-6). Furthermore, these systemic effects are associated with insulin resistance (IR), a key metabolic disorder characterized by elevated fasting insulin (FINS) levels (7). Experimental studies suggest that air pollutant-induced oxidative stress may disrupt insulin signaling pathways, resulting in hyperinsulinemia, which in turn exacerbates hepatic lipid accumulation and impairs glucose metabolism, thereby establishing a plausible mechanistic link between air pollution exposure and liver injury (8, 9).

While epidemiological studies have demonstrated a positive association between long-term exposure to PM_{2.5} and inhalable particulate matter (PM₁₀) and the incidence of diabetes as well as elevated fasting plasma glucose levels, along with a significant correlation between diabetes and non-alcoholic fatty liver disease (NAFLD) (10-12), critical knowledge gaps persist. Current research has largely concentrated on either chronic health outcomes or the effects of individual pollutants. However, the acute hepatotoxic effects of short-term exposure to realistic mixtures of air pollutants remain poorly characterized (13). This gap needs to be resolved given that humans are invariably exposed to a complex mixture of air pollutants that are often highly correlated, making it difficult to disentangle their individual effects using traditional single-pollutant models, which are susceptible to co-exposure confounding and cannot evaluate the overall mixture effect. To address this methodological challenge and provide a more realistic assessment of the combined hepatotoxicity, we employed

weighted quantile sum (WQS) regression, an approach designed to estimate the overall effect of a mixture and identify the most influential pollutants. More importantly, the potential role of FINS as a mediating biomarker in this association has not been adequately investigated.

In the course of urbanization, industry and agriculture modernization, air pollution has emerged as a particularly pressing issue in China (14). In recent years, despite the significantly improving trend in China's air quality, seasonal fluctuations persist in certain regions. Due to its unique geographical environment and climatic conditions, Xinxiang remains one of the most polluted cities in the Central Plains Urban Agglomeration of China (15). Moreover, the study area encompasses not only residential and agricultural regions but also extensive industrial facilities (e.g., chemical and manufacturing plants) and transportation infrastructure (e.g., Highway G4 and National Highway 107). Consequently, the residential population is exposed to a complex mixture of air pollutants originating from both industrial and traffic-related sources, making this setting particularly suitable for the present study.

Given the liver's susceptibility to damages from multiple environmental factors including air pollution, biomarkers of liver function may be feasible to assess the complex exposure to air pollutants (16, 17). Thus far, there is limited evidence to demonstrate the association between air pollution and the liver function of rural residents.

This present study relied on a prospective cohort of chronic diseases in Henan Province to construct models of exposure effects of multiple air pollutants on liver function for the first time in the suburb of Xinxiang, Central China (18). By integrating multiple statistical analysis methods, this research sought to realize two primary objectives: 1) characterization of the association of criteria air pollutants including PM_{2.5}, PM₁₀, NO₂, sulfur dioxide (SO₂), O₃, and carbon monoxide (CO), and liver function biomarkers including aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (ALP), indirect bilirubin (IBIL), direct bilirubin (DBIL), and total bilirubin (TBIL); 2) determination of the mediating role and contribution of FINS to air pollutant associated liver dysfunction. This study was

expected to provide evidence for assessment of air pollution-associated liver toxicity and insight into the early enzymatic alterations of liver in rural residents.

Methods

1. Study design and participant enrollment

This research was based on the Henan Rural Cohort (2016YFC0900803) in Xinxiang, Central China. The baseline surveys were conducted from April to June 2017 in multiple administrative villages selected through cluster random sampling in Xinxiang followed by follow-up surveys carried out from May to July 2021. A total of 5,133 adults participated in both surveys, and their information was collected via questionnaires, physical examinations, and biological specimen collection (18). The participants with the following conditions were excluded from this study: 1) presence of tumor, liver disease, chronic hepatitis, or gallbladder disease at the time of the baseline survey; 2) absence of a recorded survey date; and 3) lack of key data such as liver function biomarkers and blood pressure (**Additional file 1: Fig. S1**). As the analysis was retrospective, a prior sample-size calculation was not applied in this present study. However, a post-hoc power calculation based on the observed standard deviations and the published effect size (2.05 % increase in ALT per 10 $\mu\text{g}/\text{m}^3$ $\text{PM}_{2.5}$) showed 87 % power ($\alpha = 0.05$, two-sided), confirming that the present sample (N=4092) was sufficient to detect the association (19).

Before enrolled in this study, the participants gave written informed consent. The study protocol was reviewed and approved by the Ethics Committee of Xinxiang Medical University (approval number: XYLL2016242), which was in line with the guidelines stipulated in the Declaration of Helsinki.

2. Collection and examinations of biological specimens

Medical staff collected fasting blood samples (after at least 8 hours of fasting) from the antecubital vein of each participant in strict accordance with aseptic techniques. Within one hour of collection, the samples were centrifuged, separated, and aliquoted, then transported on the same day to a certified laboratory via cold chain for biochemical analysis. Prior to measurement, all instruments were calibrated to ensure analytical reliability. The aliquoted samples were subsequently stored at –

80 °C to prevent repeated freeze-thaw cycles. Thereafter, bilirubin fractions (IBIL, DBIL, and TBIL), liver enzymes (AST, ALT, and ALP), lipids [total cholesterol (TC), triglycerides (TG), low-density lipoprotein cholesterol (LDL-C), and high-density lipoprotein cholesterol (HDL-C)], and fasting insulin (FINS) were measured with a fully automatic biochemical analyzer.

3. Definition and classification of covariates

The covariates adjusted in the models were selected referred to previously published methods (20, 21) and expert domain knowledge. Age (years), body mass index (BMI, kg/m²), temperature (°C), and relative humidity (%) were treated as continuous variables. Daily average temperature and relative humidity were retrieved from the China Meteorological Data Information Network (<https://data.cma.cn>). Sex (male/female), smoking status (never/ former / current), drinking status (never/ former / current), education levels (elementary school or below/ middle school/ high school or above), average monthly income (< 500 RMB/ 500 - 999 RMB/ ≥ 1000 RMB), hypertension (yes/no), and dyslipidemia (yes/no) were specified as categorical variables. Hypertension was defined by one of the following criteria: 1) a self-reported history of hypertension diagnosed at a secondary-level hospital or above and the use of antihypertensive medication within the past two weeks; 2) systolic blood pressure ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg (22). Dyslipidemia was defined by any of the following: 1) a self-reported history of dyslipidemia diagnosed at a secondary-level hospital or above and treatment with lipid-lowering drugs within the past two weeks; 2) TC ≥ 5.2 mmol/L; 3) TG ≥ 1.7 mmol/L; 4) LDL-C ≥ 3.4 mmol/L; 5) HDL-C < 1.0 mmol/L (23).

4. Air pollution exposure assessment

The daily average concentrations of air pollutants (PM_{2.5}, PM₁₀, NO₂, SO₂, O₃, and CO) were obtained from the Yangtze River Delta Science Data Center, National Earth System Science Data Sharing Infrastructure, National Science & Technology Infrastructure of China (<http://geodata.nnu.edu.cn/>). These datasets were generated using a spatiotemporal extreme random tree (ST-ERT) model that integrated ground-based monitoring data, atmospheric reanalysis data, and emission inventory

information. Model predictions showed strong agreement with observed measurements from 128 ground monitoring stations and successfully passed 10-fold cross-validation, with cross-validated R^2 values ranging from 0.80 to 0.93. The spatial resolution was $1 \text{ km} \times 1 \text{ km}$ for the 2021 air pollutant datasets and the 2017 $\text{PM}_{2.5}$, PM_{10} , and O_3 datasets, and $10 \text{ km} \times 10 \text{ km}$ for the 2017 NO_2 , SO_2 , and CO datasets. Daily pollutant concentrations were extracted using Geographic Information System software and linked to participants' residential addresses. To investigate the temporal pattern of exposure–health associations, we evaluated both single-day lagged exposures (lag0 to lag7) and moving average cumulative lagged exposures (lag01 to lag07).

5. Statistical analysis

Baseline demographic characteristics, air pollutant concentrations, and liver function biomarkers were presented as means $\pm$ standard deviations or frequencies (%). The Spearman rank correlation test was employed to assess the correlations between air pollutants and meteorological factors.

5.1 Statistical model 1: linear mixed-effects (LME) model

The associations between individual air pollutant and liver function biomarker were evaluated using LME model. The levels of liver function biomarkers typically exhibited an approximately normal distribution following natural logarithmic transformation. The percentage change $\{[\exp(\beta \times \text{IQR}) - 1] \times 100\}$ and 95% confidence interval (CI) of levels of liver function biomarkers were calculated for each IQR increase in air pollutant levels, where β denoted the regression coefficient on a logarithmic scale. Model 1 served as a crude model. Model 2 was adjusted for age, sex, BMI, and natural cubic spline fits for temperature and humidity. Model 3 was further regulated for potentially risky lifestyle factors (smoking, alcohol status) and sociodemographic characteristics (education level, average monthly income). Model 4 was additionally controlled for hypertension and hyperlipidemia.

5.2 Statistical model 2: WQS regression model

WQS regression models were employed to assess the combined effect of the air pollutant mixture of $\text{PM}_{2.5}$, PM_{10} , NO_2 , SO_2 , CO and O_3 on liver function, addressing

challenges posed by multicollinearity among air pollutants. To ensure comparability, all pollutant concentrations were first converted to a uniform unit of $\mu\text{g}/\text{m}^3$. For the WQS model itself, these values were then transformed into deciles (1-10), which is the standard approach to normalize distributions and reduce outlier influence. The analysis was conducted under a pre-specified positive directionality assumption, informed by our preliminary single-pollutant analyses. Air pollutant concentrations were transformed into deciles, and a WQS index was constructed through 1000 bootstrap iterations using a 40%/60% training-validation split. A significant positive association of the WQS index indicated a joint adverse effect. The relative contribution of each air pollutant was determined by its respective weight, the numbers in parentheses, within the significant index.

5.3 Statistical model 3: mediation effect model

To assess the impact of short-term exposure to air pollutants on liver function biomarker levels and the potential mediating role of FINS, we applied a causal mediation analysis within the counterfactual framework. The analysis was implemented using three LMEs to account for within-subject correlations arising from repeated measurements. First, the total effect model estimated the overall association between the air pollutant and the liver function biomarker. Second, the mediator model estimated the association between the air pollutant and FINS. Finally, the direct effect model estimated the association between the air pollutant and the liver function biomarkers after adjusting for FINS. All models were adjusted for a comprehensive set of covariates considered potential confounders based on prior evidence, including sex, age, BMI, smoking status, alcohol consumption, education level, monthly income, and history of hypertension and dyslipidemia, in an attempt to meet the critical assumptions of no unmeasured confounding. Furthermore, natural cubic spline terms (with 3 degrees of freedom) for ambient temperature and relative humidity were included to control for potential nonlinear meteorological confounding. The temporal order of exposure $\rightarrow$ mediator $\rightarrow$ outcome was supported by the study design. The statistical significance of the indirect effect was assessed using a non-parametric bootstrap method with 1000 resamples to obtain robust confidence intervals.

5.4 Stratified analysis and sensitivity analysis

To investigate whether demographic and lifestyle factors modified the effects, stratified analyses were conducted based on age (< 60 years, ≥ 60 years), sex (male, female), smoking status (yes, no), and alcohol consumption (yes, no). Finally, to evaluate the robustness of our primary findings, we conducted a series of sensitivity analyses: 1) Age was modeled using a restricted cubic spline function with three knots to account for potential non-linear associations with liver function biomarkers, replacing the linear age term in the model; 2) To minimize confounding from major comorbid conditions, we excluded participants diagnosed with tumors, liver diseases, chronic hepatitis, or gallbladder diseases during the follow-up period; 3) We further excluded observations where ALT/AST levels exceeded three times the upper limit of normal, ensuring that results were not unduly influenced by severe acute liver injury; 4) Although our study population was only surveyed during spring and summer, we further adjusted for the specific season (spring vs. summer) to account for any residual seasonal confounding within this time frame; 5) To address the potential for residual confounding by weekly activity patterns (day-of-the-week effect), we incorporated a categorical variable for the day of the week (Monday through Sunday) into the primary model.

The statistical analyses were conducted using R software (version 4.4.0) and SPSS software (version 27.0), with a significance level set at 0.05. Given the large number of statistical tests performed across single-pollutant models, mixture analyses, and mediation analyses, the potential for false positive findings was a concern. To address this, we applied the Benjamini-Hochberg (BH) procedure to control the False Discovery Rate (FDR) at a level of 0.05. This method was applied to the P -values from all primary models. Associations with an FDR-adjusted P -value of < 0.05 were considered statistically significant.

Results

1. Basic characteristics of the participants

A total of 4,096 individuals were included in the analyses in this study

(**Additional file 1: Fig. S1**), with their baseline demographic characteristics summarized in **Table 1**. The mean age of the participants was 54.29 ± 12.31 years, and 62.67% were female. The educational attainment (40.67% completed primary school or below) and economic status (71.62% had a monthly income of less than 1000 RMB) are representative of typical rural areas in Central China. Notably, 74.88% of the participants were non-smokers, and 77.34% did not consume alcohol. However, hypertension and dyslipidemia affected 45.43% and 69.24% of the subjects, respectively. In addition, the mean BMI was 25.55 ± 3.51 kg/m², indicating that the many subjects were overweight.

Table 2 shows the dynamic changes in air pollutant concentrations and the alterations of liver function biomarkers across the two surveys. During the baseline period (2017), the concentrations of PM_{2.5} (50.86 ± 9.55 µg/m³), PM₁₀ (125.01 ± 34.52 µg/m³), and O₃ (176.15 ± 31.68 µg/m³) exceeded the 2021 WHO global air quality guidelines (15µg/m³, 45µg/m³; 100µg/m³), respectively. During the follow-up period (in 2021), the average concentration of PM_{2.5} was 28.14 ± 5.75 µg/m³, showing a significant decrease compared to those in 2017; the PM₁₀ concentration was 80.59 ± 30.47 µg/m³, also showing a downward trend. However, the O₃ level rose to 181.93 ± 21.20 µg/m³. The levels of hepatic enzymes and bilirubin spectrum demonstrated an obvious decrease compared to those in 2017. **Fig. 1** uncovered a significant correlation between air pollutants and meteorological factors.

2. Association of air pollutant exposures with biomarker of liver function

Among model 1 to model 4, only the latter demonstrated robust performance. Consequently, all subsequent analyses were based on model 4 (**Additional file 1: Fig. S2 - S4, Fig. 2** and **Additional file 1: Table S1**).

Our results revealed that short-term exposure to various air pollutants was significantly associated with alterations in multiple liver function biomarkers. The most pronounced effects were observed for bilirubin levels (IBIL, DBIL, and TBIL), which showed strong positive associations with SO₂, NO₂, and O₃, with percentage increases per IQR often exceeding 20% and reaching up to 47.66% (95%CI: 43.49%,51.95%) for IBIL with SO₂ (lag01). For the enzymes AST, ALT, and ALP, the

associations were generally positive but more modest in magnitude. PM_{2.5}, NO₂, and CO were consistently associated with increases in AST and ALP. A notable exception was the significant negative association observed between O₃ (lag5) and ALT levels (-2.17% [95%CI: 3.2%, 1.13%] per IQR increase).

In summary, our analysis demonstrated that short-term exposure to major air pollutants was significantly associated with changes in liver function biomarkers, with particularly strong effects on bilirubin metabolism. The complete set of effect estimates, CIs, and lag periods for each pollutant-biomarker pair was provided in **Additional file 1: Table S1**.

3. Association of combined exposure to air pollutants with biomarkers of liver function in the WQS model

The WQS model showed the air pollutants' contribution to alterations of liver enzymes (**Fig. 3** and **Additional file 1: Table S2**). The ALP abnormalities were mainly attributed to SO₂ (0.44), CO (0.42) and NO₂ (0.12) ($\beta = 0.012$, $P < 0.001$) exposures (**Fig. 3A**). The increases in AST levels were mostly influenced by SO₂ (0.32) CO (0.30), O₃ (0.28), and NO₂ (0.08), ($\beta = 0.017$, $P < 0.001$) (**Fig. 3B**). Notably, both NO₂ and SO₂ contributed to bilirubin metabolism effects (all $P < 0.001$) (**Fig. 3D-F**). For instance, elevated TBIL levels were strongly associated with combined exposure to SO₂ (0.62), NO₂ (0.13), and O₃ (0.09) ($\beta = 0.07$, $P < 0.001$) (**Fig. 3F**).

4. Mediation analyses

The mediation analyses revealed a complex interplay between air pollution exposure, FINS, and liver function biomarkers, with distinct patterns depending on air pollutant types and hepatic endpoints (**Table 3**). Six air pollutants consistently exhibited significant mediation through FINS for bilirubin markers (IBIL, DBIL, TBIL). Specifically, PM_{2.5} exposure increased TBIL levels via FINS (indirect effect: 8.66×10^{-4} , 95% CI: 7.81×10^{-4} , 1.25×10^{-3} ; mediation proportion: 16.03%, 95% CI: 14.35%, 25.35%). FINS showed the strongly mediation between NO₂ exposure and changes of IBIL (11.41%, 95% CI: 9.78%, 16.28%), DBIL (5.32%, 95% CI: 5.02%, 8.47%) and TBIL (11.56%, 95% CI: 9.97%, 16.58%). Interestingly, our findings from

this study suggested that FINS might potentially exert an inhibitory effect on the association between air pollutants and liver enzyme levels. FINS reduced the influence of PM_{2.5}, PM₁₀, NO₂, and SO₂ on ALT by 49.92% (95% CI: -96.56%, -29.99%), 24.65% (95% CI: -44.24%, -15.26%), 61.97% (95% CI: -143.92%, -27.34%), and 59.16% (95% CI: -140.95%, -31.15%), NO₂ and SO₂ on ALP by 4.79% (95% CI: -9.55%, -0.25%) and 4.49% (95% CI: -8.59%, -0.71%).

5. Stratified analyses

Stratified analyses revealed that individual characteristics of the subjects significantly modified the association of air pollutants with liver function markers (**Fig. 4**). For example, the impact of PM_{2.5}, PM₁₀, SO₂, NO₂, or CO on DBIL was significantly greater in women than in men. For each IQR change in PM_{2.5}, PM₁₀, SO₂, NO₂, and CO, the percentage changes in DBIL were 7.61% (95% CI: 5.89%, 9.35%), 3.96% (95% CI: 2.23%, 5.71%), 28.69% (95% CI: 22.73%, 34.94%), 20.54% (95% CI: 17.84%, 23.31%), and 14.60% (95% CI: 12.23%, 17.02%), respectively, in male, but were 10.78% (95% CI: 9.36%, 12.23%), 4.44% (95% CI: 3.09%, 5.81%), 37.40% (95% CI: 32.66%, 42.32%), 26.05% (95% CI: 23.86%, 28.28%), and 20.08% (95% CI: 18.09%, 22.11%) in female. Furthermore, there were similar effect changes of DBIL levels in individuals under the age of 60, without a history of alcohol consumption, and non-smokers. However, ALP levels were significantly affected by air pollutant exposure among individuals aged ≥ 60 years, males, or those with a history of smoking. Exposure to PM_{2.5}, SO₂, NO₂, and CO was associated with significant increases in ALP levels among smokers, with corresponding percentage changes of 5.12% (95% CI: 3.64%, 6.61%), 11.14% (95% CI: 8.69%, 13.64%), 8.94% (95% CI: 6.92%, 11.00%), and 7.81% (95% CI: 6.01%, 9.64%), respectively.

6. Sensitivity analyses

Sensitivity analyses were conducted to evaluate the robustness of the results from the primary model. The non-linear effect test for age showed the stability of the preliminary results. The association between concentrations of PM_{2.5} (lag3) and ALT remained stable, with the effect estimate decreasing slightly from 2.74% (95%CI: 1.58%, 3.91%) to 2.62% (95%CI: 1.48%, 3.80%) (**Additional file 1: Fig. S5** and

Additional file 1: Table S3). To minimize potential confounding effects on liver function due to disease progression, this study excluded the participants with newly diagnosed tumors, liver diseases, chronic hepatitis, or gallbladder diseases (n=395) during follow-up examination in 2021. The results demonstrated that the associations between air pollutants and liver function biomarkers were consistent with the main analyses (**Additional file 1: Fig. S6** and **Additional file 1: Table S4**). Furthermore, after excluding cases of severe liver injury (n=6), the associated changes were negligible (**Additional file 1: Fig. S7** and **Additional file 1: Table S5**). After further adjustment for seasonality, the effect estimate remained robust. Each IQR increment in PM_{2.5} was associated with a 3.48% (95% CI: 2.61%, 4.44%) increase in ALT levels, with minimal attenuation (<10%) relative to the primary model estimate (**Additional file 1: Fig. S8** and **Additional file 1: Table S6**). In a sensitivity analysis adjusting for the day of the week, the effect estimates for the associations between air pollutants and liver function biomarkers exhibited greater stability across different lag periods compared to the main model (**Additional file 1: Fig. S9** and **Additional file 1: Table S7**).

Discussion

NAFLD, the most common liver disease in China, and its association with environmental stress have become a key research issue in hepatology. Although an increasing amount of evidence has associated long-term exposure to air pollutants with NAFLD, the relationship between short-term exposure and liver function remains under-explored (24, 25). So far, only a few studies have examined the mediating effect of FINS on the relationship between air pollution and liver function in rural populations. The findings of this study demonstrated that short-term exposure to air pollutants was linked to changes in liver function biomarkers, with FINS exerting a partial mediating effect.

A systematic review and meta-analysis showed that each 10 µg/m³ rise in PM_{2.5} concentration corresponds to a 4.45% increase in ALT levels, and a 3.99% increase in AST levels (26). In a cohort study of HIV/AIDS patients, a 10 µg/m³ increase in

PM_{2.5} levels was positively associated with the percent change of AST concentrations from 1.92% (95%CI: 3.13, 4.38) to 6.09% (95%CI: 9.25, 12.38) (27). In addition, acute O₃ exposure is also linked to elevated concentrations of AST, and ALP (27). These results were consistent with the findings from this present study, showing a consistent and significant positive correlation between criteria air pollutants and six critical hepatic enzymes. Moreover, this study demonstrated a positive association between air pollution with blood levels of bilirubin species, including IBIL, DBIL and TBIL in rural residents, similar to the results from a previous cross-sectional study demonstrating a positive correlation between PM_{2.5}, O₃, NO₂, and TBIL levels (13). However, our study revealed a distinct association pattern between short-term exposure to O₃ and ALT levels compared to other air pollutants. While exposures to PM_{2.5}, PM₁₀, NO₂, and SO₂, were generally associated with elevated ALT levels, suggesting potential hepatotoxic effects, O₃ exposure was consistently associated with reduced ALT levels across multiple single-day and moving average lag periods. To our knowledge, the inverse association between O₃ and ALT levels represents a novel finding. The discrepancy between our results and those reported in previous studies may be attributed to differences in study population characteristics (such as rural residents with distinct exposure profiles and lifestyle patterns), exposure conditions, including moderate O₃ levels combined with co-exposure to other pollutants, or variations in the statistical modeling of lag structures. Noteworthy, O₃ has been shown to present dual immunoregulatory characteristics, acting as both an immune activator and suppressor depending on its concentration and context (28, 29). These findings underscored the complexity of air pollutant effects.

Currently, limited research has been conducted on the relationship between combined exposure to air pollutants and alterations in hepatic enzymes and the bilirubin profile. In this study, the WQS regression was employed to systematically analyze the synergistic toxicity of six major air pollutants on biomarkers of liver function. Our WQS regression analysis revealed that the hepatotoxicity of air pollution is driven by endpoint-specific mixture effects. SO₂ and NO₂ emerged as the primary contributors to hepatocellular injury, as indicated by elevated levels of liver

enzymes, and were also associated with disrupted bilirubin metabolism. Although SO₂ and NO₂ were not the most abundant pollutants in this cohort (**Table 2**), this does not imply that they are inherently the most toxic. The prominent contribution of SO₂ and NO₂ to the WQS index reflected their greater biological relevance to the specific liver outcomes examined in this study, rather than indicating a universally higher toxicity compared to other pollutants. Furthermore, PM₁₀ exhibited heightened impact on ALT elevation. These findings challenged single-pollutant paradigms and advocated for integrated control strategies targeting multipollutant emissions, particularly SO₂ and NO₂ from fossil fuel combustion, to mitigate distinct pathways of liver injury. While constrained by model assumptions, our study highlighted the importance of air pollutant interactions in environmental health risk assessment.

This study uncovered the dual roles of FINS in the relationship between air pollution and disorders of bilirubin metabolism, as well as liver enzyme levels, showing that air pollutant levels were significantly associated with the levels of bilirubins and mediated by FINS. Once entering the human body through the airways, air pollutants induce oxidative stress, activate inflammatory pathways, and interfere with insulin signaling and alter FINS levels (30-33). As a critical indicator of insulin resistance, abnormal FINS further impacts the metabolic regulation of bilirubin in the liver (34). Insulin regulates the uptake, binding, and excretion of bilirubin by modulating transporter and enzymatic activities (35, 36), which could be disrupted by changes of FINS levels. Bilirubin, an endogenous antioxidant, can exacerbate air pollutant-induced oxidative damage when its metabolism is disrupted (37, 38). Notably, FINS is proposed to exhibit a protective modulatory effect on the relationship between air pollution and liver enzymatic (ALT, ALP) disorders in this study. Despite PM_{2.5} and other pollutants increasing liver enzyme metabolism levels through oxidative stress and inflammation, changes of FINS may trigger adaptive protective mechanisms, such as enhancing the expression of antioxidant enzymes, including superoxide dismutase and glutathione peroxidase, and inhibiting pro-inflammatory factor release, thus protecting liver function (39). These dual roles indicate that FINS not only mediates air pollutant-induced hepatic pathogenesis but

also presents the potential as therapeutic target. Future researches are warranted to include prospective cohort and experimental studies to confirm causal associations.

Furthermore, stratified analyses unraveled significant population heterogeneity in the impact of air pollutants on DBIL. Particularly, the response of DBIL was more pronounced among the young, female, non-smoking, and non-drinking populations. However, documented research has mainly focused on middle-aged and elderly individuals as well as newborns. In the young group, DBIL levels have significantly increased. It is assumed that this may be due to excessive metabolism and intense exposure to environmental insults. Previous studies have demonstrated that DBIL serves as an endogenous antioxidant, and a mild increase in DBIL can exert protective effects (40, 41). The antioxidant system in individuals not exposed to tobacco and alcohol remains in a basal state (42). Upon exposure to air pollution, DBIL, as a rapidly mobilized endogenous antioxidant, increases preferentially (43, 44). In addition, estrogen in women prolongs the retention time of DBIL in the bloodstream, enhancing its antioxidant capacity (45). Smoking has been shown to be associated with increased serum ALP levels, and air pollution may further intensify this effect (46). However, additional studies are required to clarify the underlying mechanisms.

Several limitations warrant consideration for future research. The GIS-based static exposure assessment inadequately captured the spatiotemporal variability of participants' daily activity patterns, potentially introducing exposure measurement error that may have compromised the validity of effect estimates. In addition, animal studies are needed to confirm whether air pollution affects liver function through FINS. Importantly, unmeasured confounding due to lifestyle factors such as sleep quality and dietary intake cannot be ruled out; although these variables may bias the observed associations, their absence should be acknowledged as a key limitation that underscores the need for future data collection and adjustment. Additionally, WQS regression assumes a linear and additive relationship between air pollutant mixture and health outcomes. Although our preliminary analyses using LMEs supported the presence of a linear association, the potential for more complex, non-linear relationships cannot be excluded, as these may not be adequately captured by the

WQS model. Future studies with larger sample sizes should consider employing more flexible analytical methods, such as Bayesian Kernel Machine Regression (BKMR), to fully characterize potential non-linear dose-response patterns and interactions among air pollutants. Our assessment of lag effects using single-day lags and moving averages, while informative and less model-dependent, does not provide a smoothed lag-response curve like a Distributed Lag Model (DLNM) would. Future research with larger longitudinal data could apply DLNM to further refine the understanding of the temporal dynamics observed here. Finally, although the study periods (April–June 2017 and May–July 2021) overlapped with the COVID-19 pandemic, all field visits were conducted during local post-lockdown phases when restrictions had been lifted and residents had resumed normal daily activities. Thus, participation rates and mobility-related exposure patterns were likely comparable to pre-pandemic conditions. Nevertheless, we cannot entirely exclude the possibility that residual changes in traffic volume, industrial emissions, or personal time–activity patterns during the early recovery phase may have altered true exposure levels, leading to non-differential misclassification that would bias effect estimates toward the null.

Conclusions

This study identifies significant positive associations between short-term exposure to air pollutants and biomarkers of liver function in a rural cohort study, in which FINS may act as a key mediator in the pathways of air pollution-induced liver abnormality. These findings highlight FINS as a potential biomarker for early intervention in air pollution-related metabolic liver damage and support the integration of liver health protection into public health strategies targeting the control of airborne pollutants, particularly SO₂ and NO₂.

List of abbreviations

WQS: weighted quantile sum

FINS: fasting insulin

PM_{2.5}: fine particulate matter

AST: aspartate aminotransferase

CI: confidence interval

TBIL: total bilirubin

IBIL: indirect bilirubin

DBIL: direct bilirubin

NO₂: nitrogen dioxide

O₃: ozone

IR: insulin resistance

NAFLD: non-alcoholic fatty liver disease

PM₁₀: inhalable particulate matter

SO₂: sulfur dioxide

ALT: alanine aminotransferase

ALP: alkaline phosphatase

TC: total cholesterol

TG: triglycerides

LDL-C: low-density lipoprotein cholesterol

HDL-C: and high-density lipoprotein cholesterol (HDL-C)

BMI: body mass index

LME: linear mixed-effects model

IQR: interquartile range

BKMR: Bayesian Kernel Machine Regression

DLNM: Distributed Lag Model

Declarations

Acknowledgements: We thank the medical staff for their technical support and resident volunteers from local communities.

Consent for Publication: Written informed consent was obtained from all individual participants included in the study. The consent form explicitly stated that their

anonymized data could be used for publication in scientific journals.

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Data Availability: The data obtained from the current study can be accessed from the corresponding author upon reasonable request.

Authors' Contributions: All authors read and approved the final manuscript.

MW: Methodology, Data curation, Validation, Visualization, Software, Formal analysis, writing-original draft.

AZ: Methodology, Data curation, Software, Validation, Visualization.

SZ: Methodology, Software, Validation, Visualization.

YZ: Methodology, Software, Validation.

HS: Methodology, Validation.

QW: Methodology, Validation.

JL: Methodology, Validation.

YD: Methodology, Validation.

YC: Methodology, Validation.

JS: Methodology, Validation.

LZ: Methodology, Validation.

HW: Data curation.

WW: Conceptualization, Funding acquisition, Supervision, Writing - review & editing, Project administration, Resources, Validation.

Competing interests: The authors declare that they have no competing interests

Ethics Approval and Consent to Participate: The study protocol was reviewed and approved by the Ethics Committee of Xinxiang Medical University (approval number: XYLL2016242), which was in line with the guidelines stipulated in the Declaration of

Helsinki. All participants consented to participate this study.

Additional File Description:

Additional File 1: Figs. S1-S9.

Fig. S1- [Participant selection procedure for this study].

Fig. S2- [Association between short-term exposure to air pollutants and levels of liver function biomarkers (Model 1)].

Fig. S3- [Association between short-term exposure to air pollutants and levels of liver function biomarkers (Model 2)].

Fig. S4- [Association between short-term exposure to air pollutants and levels of liver function biomarkers (Model 3)].

Fig. S5- [Non-linear effect of age on the association between air pollutants and liver function biomarkers using restricted cubic splines with three degrees of freedom].

Fig. S6- [Sensitivity analyses excluding individuals diagnosed with diseases during follow-up to assess the impact on liver function biomarkers].

Fig. S7- [Sensitivity analyses excluding samples with ALT/AST >3 times the upper limit to assess the impact of extreme liver injury].

Fig. S8- [After seasonal adjustment, the association between air pollutants and liver function biomarker.].

Fig. S9- [After adjusting for the day of the week, the association between air pollutants and liver function biomarkers].

Additional File 1: Tables S1-S6.

Table S1- [Association between short-term exposure to air pollutants and levels of ALP(A), ALT(B), AST(C), IBIL(D), DBIL(E), and TBIL(F) in model 4].

Table S2- [The WQS index of liver function biomarkers].

Table S3- [Association between short-term exposure to air pollutants and levels of ALP(A), ALT(B), AST(C), IBIL(D), DBIL(E), and TBIL(F) in the sensitivity analysis model 1].

Table S4- [Association between short-term exposure to air pollutants and levels of

ALP(A), ALT(B), AST(C), IBIL(D), DBIL(E), and TBIL(F) in the sensitivity analysis model 2].

Table S5- [Association between short-term exposure to air pollutants and levels of ALP(A), ALT(B), AST(C), IBIL(D), DBIL(E), and TBIL(F) in the sensitivity analysis model 3].

Table S6- [Association between short-term exposure to air pollutants and levels of ALP(A), ALT(B), AST(C), IBIL(D), DBIL(E), and TBIL(F) in the sensitivity analysis model 4].

Table S7- [Association between short-term exposure to air pollutants and levels of ALP(A), ALT(B), AST(C), IBIL(D), DBIL(E), and TBIL(F) in the sensitivity analysis model 5].

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Table 1. Characteristics of participant demographics, lifestyle factors, and medical history.

Characteristics	Mean $\pm$ SD /n (%)
(N = 4096)	
Age (years)	54.29 $\pm$ 12.31
Sex	
male	1529 (37.33)
female	2567 (62.67)
Education lever	
Elementary school or below	1666 (40.67)
Junior high school	1590 (38.82)
High school and above	840 (20.51)
Monthly income	
<500 RMB	1472 (35.93)
500-999 RMB	1462 (35.69)
$\geq$ 1000 RMB	1162 (28.37)
Smoking status	
Never	3067 (74.88)
Former	731 (17.85)
Current	298 (7.28)
Drinking status	
Never	3168 (77.34)
Former	773 (18.87)
Current	155 (3.78)
BMI	
Under weight	44 (1.07)
Normal weight	1364 (33.30)
Over weight	1765 (43.09)
Obesity	923 (22.53)
Hypertension	
No	2235 (54.57)
Yes	1861 (45.43)
Dyslipidemia	
No	1260 (30.76)
Yes	2836 (69.24)

Table 2. Mean concentrations of air pollutants, meteorological factors, and liver function biomarkers of the participants during the 7 days before baseline and follow-up surveys.

2017 year

2021 year

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PM_{2.5}, particulate matter with an aerodynamic diameter ≤ 2.5 μm ; PM₁₀, particulate matter with an aerodynamic diameter ≤ 10 μm ; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; T, temperature; RH, relative humidity; AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; IBIL, indirect bilirubin; DBIL, direct bilirubin; TBIL, total bilirubin.

	Mean $\pm$ SD	Percentile			Mean $\pm$ SD	Percentile		
		25 th	50 th	75 th		25 th	50 th	75 th
PM _{2.5} , $\mu\text{g}/\text{m}^3$	50.86 $\pm$ 9.55	44.81	45.69	54.26	28.14 $\pm$ 5.75	22.86	27.96	33.51
PM ₁₀ , $\mu\text{g}/\text{m}^3$	125.01 $\pm$ 34.52	99.21	114.94	128.77	80.59 $\pm$ 30.47	55.88	65.85	106.48
SO ₂ , $\mu\text{g}/\text{m}^3$	25.61 $\pm$ 4.10	22.20	26.44	28.51	9.33 $\pm$ 2.12	6.99	9.31	11.69
NO ₂ , $\mu\text{g}/\text{m}^3$	39.55 $\pm$ 2.42	37.69	39.34	41.04	22.93 $\pm$ 4.80	18.12	24.75	26.69
O ₃ , $\mu\text{g}/\text{m}^3$	176.15 $\pm$ 31.68	149.96	169.08	202.33	181.93 $\pm$ 21.20	163.03	186.03	195.65
CO, mg/m ³	1.00 $\pm$ 0.05	0.96	0.99	1.04	0.66 $\pm$ 0.08	0.60	0.66	0.71
T, °C	23.88 $\pm$ 2.58	21.70	24.15	25.70	28.71 $\pm$ 1.42	27.73	28.63	29.91
RH, %	46.95 $\pm$ 8.21	41.84	46.13	51.72	48.70 $\pm$ 14.28	34.52	47.66	63.63
AST, U/L	23.37 $\pm$ 12.54	19.00	22.00	25.00	22.84 $\pm$ 10.68	18.00	21.00	25.00
ALT, U/L	22.37 $\pm$ 21.79	14.00	18.00	25.00	21.37 $\pm$ 15.07	14.00	18.00	24.00
ALP, U/L	86.61 $\pm$ 25.31	70.00	84.00	101.00	79.98 $\pm$ 24.59	63.00	77.00	94.00
IBIL, $\mu\text{mol}/\text{L}$	12.43 $\pm$ 5.87	8.40	11.40	15.18	8.63 $\pm$ 4.34	5.60	7.70	10.50
DBIL, $\mu\text{mol}/\text{L}$	4.02 $\pm$ 1.67	2.97	3.70	4.70	3.02 $\pm$ 1.66	2.20	2.70	3.50
TBIL, $\mu\text{mol}/\text{L}$	16.45 $\pm$ 7.27	11.50	15.10	19.80	11.65 $\pm$ 5.31	8.10	10.50	13.90

Table 3. Mediating role of fasting insulin in the association between air pollution exposure and biomarkers of liver function.

Pathways	Indirect effect (95% CI)	P^a	Mediation proportions (95% CI)	P^b
PM_{2.5}				
AST	-1.12×10^{-4} (-1.66×10^{-4} , 6.91×10^{-5})	0.40	-8.72% (-14.69%, 4.94%)	0.40
ALT	-1.12×10^{-4} (-1.28×10^{-4}, -8.27×10^{-4})	0.002	-49.92% (-96.56%, -29.99%)	0.002
ALP	-8.03×10^{-5} (-2.03×10^{-4} , 4.22×10^{-5})	0.23	3.63% (-9.75%, 1.97%)	0.23
IBIL	9.61×10^{-4} (8.53×10^{-4}, 1.37×10^{-3})	0.002	16.19% (14.04%, 24.96%)	0.002
DBIL	3.19×10^{-4} (3.00×10^{-4}, 5.53×10^{-4})	0.002	6.22% (6.18%, 11.57%)	0.002
TBIL	8.66×10^{-4} (7.81×10^{-4}, 1.25×10^{-3})	0.002	16.03% (14.35, 25.32%)	0.002
PM₁₀				
AST	-1.10×10^{-5} (-1.91×10^{-5} , 1.66×10^{-5})	0.97	-4.92% (15.76%, 12.57%)	0.98
ALT	-2.56×10^{-4} (2.90×10^{-4}, 1.61×10^{-4})	0.002	-24.65% (44.24%, 15.26%)	0.002
ALP	-6.64×10^{-7} (-2.20×10^{-5} , 2.32×10^{-5})	0.97	-2.50% (12.89%, 9.85%)	0.98
IBIL	9.32×10^{-5} (5.59×10^{-5}, 1.70×10^{-4})	0.002	9.60% (5.59%, 18.51%)	0.002
DBIL	6.74×10^{-5} (4.90×10^{-5}, 1.36×10^{-4})	0.002	8.67% (6.99%, 18.92%)	0.002
TBIL	8.54×10^{-5} (5.16×10^{-5}, 1.57×10^{-4})	0.002	10.70% (6.52%, 22.03%)	0.002
NO₂				
AST	-2.69×10^{-4} (3.75×10^{-4} , 8.10×10^{-5})	0.22	-6.77% (-10.74%, 1.97%)	0.22
ALT	-1.38×10^{-3} (-1.53×10^{-3}, -9.43×10^{-4})	0.002	-61.97% (-143.92%, -27.34%)	0.01
ALP	-2.46×10^{-4} (-4.71×10^{-4}, -1.40×10^{-5})	0.046	-4.79% (-9.55%, -0.25%)	0.046
IBIL	2.18×10^{-3} (1.87×10^{-3}, 2.97×10^{-3})	0.002	11.41% (9.78%, 16.28%)	0.002
DBIL	7.86×10^{-4} (7.46×10^{-4}, 12.71×10^{-3})	0.002	5.32% (5.02%, 8.47%)	0.002
TBIL	1.97×10^{-3} (1.69×10^{-3}, 2.72×10^{-3})	0.002	11.56% (9.97%, 16.58%)	0.002
SO₂				
AST	-2.75×10^{-4} (-3.96×10^{-4} , 1.16×10^{-4})	0.26	-7.57% (12.56%, 2.98%)	0.26
ALT	-1.75×10^{-3} (-1.99×10^{-3}, -1.25×10^{-3})	0.002	-59.16% (-140.95%, -31.15)	0.006
ALP	-2.61×10^{-4} (-4.80×10^{-4}, -4.16×10^{-5})	0.017	-4.49% (-8.59%, -0.71%)	0.017
IBIL	1.64×10^{-3} (1.40×10^{-3}, 2.23×10^{-3})	0.002	7.21% (6.14%, 9.81%)	0.002
DBIL	9.77×10^{-4} (8.56×10^{-4}, 1.64×10^{-3})	0.002	5.67% (4.85%, 9.34%)	0.002
TBIL	1.48×10^{-3} (1.27×10^{-3}, 2.03×10^{-3})	0.002	7.21% (6.24%, 9.99%)	0.002
CO				
AST	-1.35×10^{-2} (-1.85×10^{-2} , 1.54×10^{-3})	0.10	-8.52% (-12.67%, 0.85%)	0.11
ALT	-5.56×10^{-2} (-6.49×10^{-2} , -3.97×10^{-2})	0.002	130.48% (-915.01%, 726.60%)	0.18
ALP	-1.02×10^{-2} (-1.92×10^{-2}, -1.07×10^{-3})	0.031	-5.19% (-10.35%, -0.51%)	0.039
IBIL	5.92×10^{-2} (5.07×10^{-2}, 7.99×10^{-2})	0.002	9.06% (7.93%, 12.81%)	0.002
DBIL	3.19×10^{-2} (3.04×10^{-2}, 5.11×10^{-2})	0.002	6.39% (5.93%, 10.02%)	0.002
TBIL	6.15×10^{-2} (5.33×10^{-2}, 8.48×10^{-2})	0.002	10.64% (9.22%, 15.08%)	0.002
O₃				
AST	-2.95×10^{-5} (-4.36×10^{-5} , 3.42×10^{-5})	0.81	-6.36% (-15.62%, 11.68%)	0.82
ALT	3.08×10^{-5} (1.41×10^{-6} , 6.93×10^{-5})	0.06	-10.66% (-77.23%, 0.78%)	0.10
ALP	-1.37×10^{-5} (-5.46×10^{-5} , 3.45×10^{-5})	0.74	-1.70% (7.77%, 4.57%)	0.74
IBIL	5.89×10^{-4} (5.03×10^{-4}, 8.04×10^{-4})	0.002	10.51% (8.74%, 14.54%)	0.002

DBIL	2.96×10⁻⁴ (2.63×10⁻⁴, 5.01×10⁻⁴)	0.002	6.86% (5.85%, 11.23%)	0.002
TBIL	5.29×10⁻⁴ (4.55×10⁻⁴, 7.28×10⁻⁴)	0.002	10.28% (8.50%, 14.24%)	0.002

PM_{2.5}, particulate matter with an aerodynamic diameter ≤ 2.5 μm ; PM₁₀, particulate matter with an aerodynamic diameter ≤ 10 μm ; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; IBIL, indirect bilirubin; DBIL, direct bilirubin; TBIL, total bilirubin; FINS, fasting insulin.

Boldface type denoted statistical significance following Benjamini-Hochberg correction for multiple comparisons.

P^a : The indirect effects were determined based on the Benjamini-Hochberg corrected P -value.

P^b : Mediation proportions were determined based on Benjamini-Hochberg adjusted P -values.

Fig. 1

Fig. 1 Spearman correlation (coefficient) between air pollutants and meteorological variables ($P < 0.05$). PM_{2.5}, particulate matter with an aerodynamic diameter $\leq 2.5 \mu\text{m}$; PM₁₀, particulate matter with an aerodynamic diameter $\leq 10 \mu\text{m}$; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; T, temperature; RH, relative humidity.

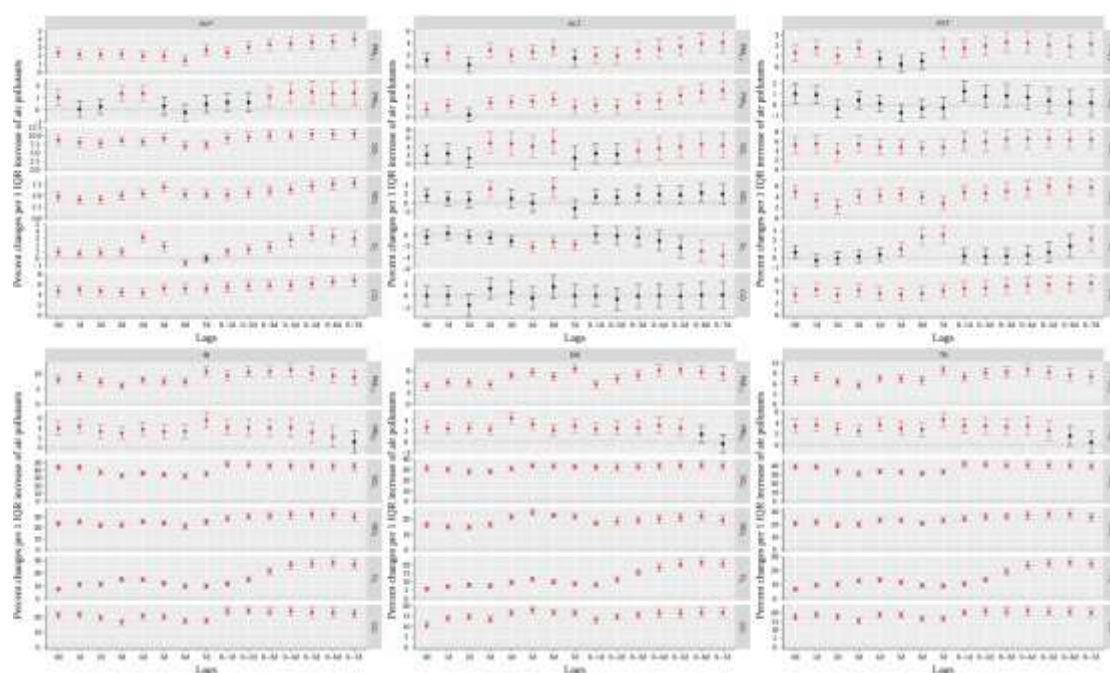
Fig. 2

Fig. 2 Association between short-term exposure to air pollutants and levels of liver function biomarkers. PM_{2.5}, particulate matter with an aerodynamic diameter $\leq 2.5 \mu\text{m}$; PM₁₀, particulate matter with an aerodynamic diameter $\leq 10 \mu\text{m}$; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; T, temperature; RH, relative humidity; AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; IBIL, indirect bilirubin; DBIL, direct bilirubin; TBIL, total bilirubin.

Error bars represent 95% confidence intervals.

Bar colors denote statistical significance following Benjamini-Hochberg correction for multiple testing: black indicates a non-significant association ($P \geq 0.05$), and red indicates a statistically significant association ($P < 0.05$).

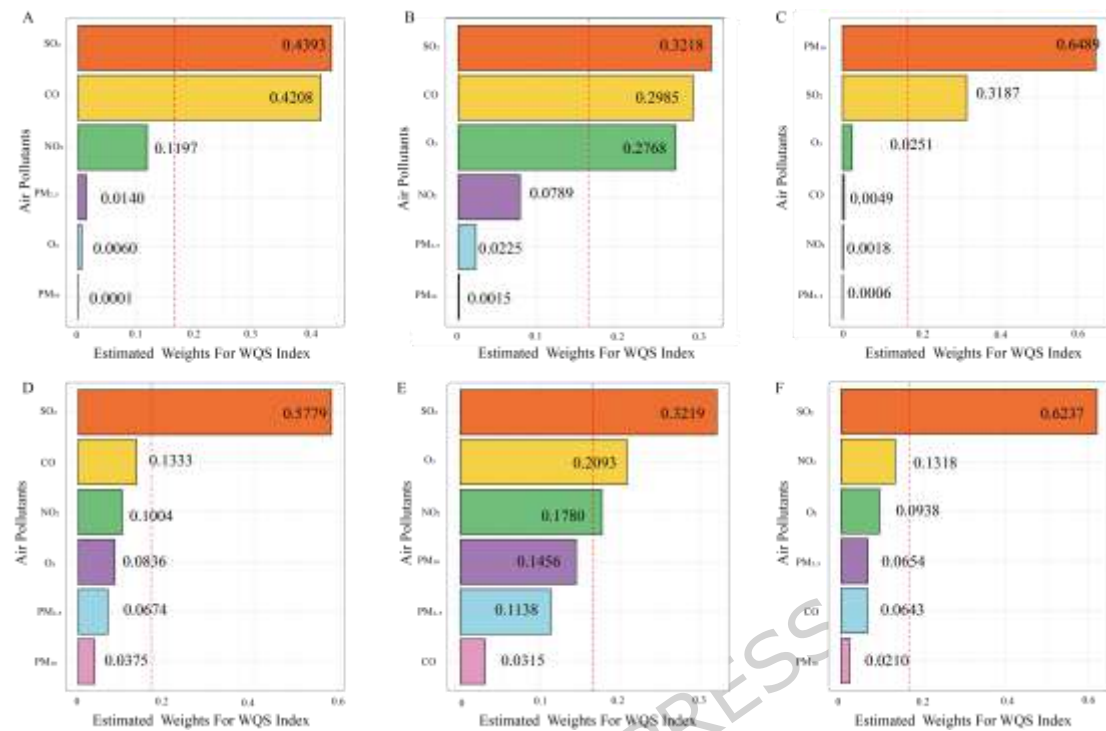
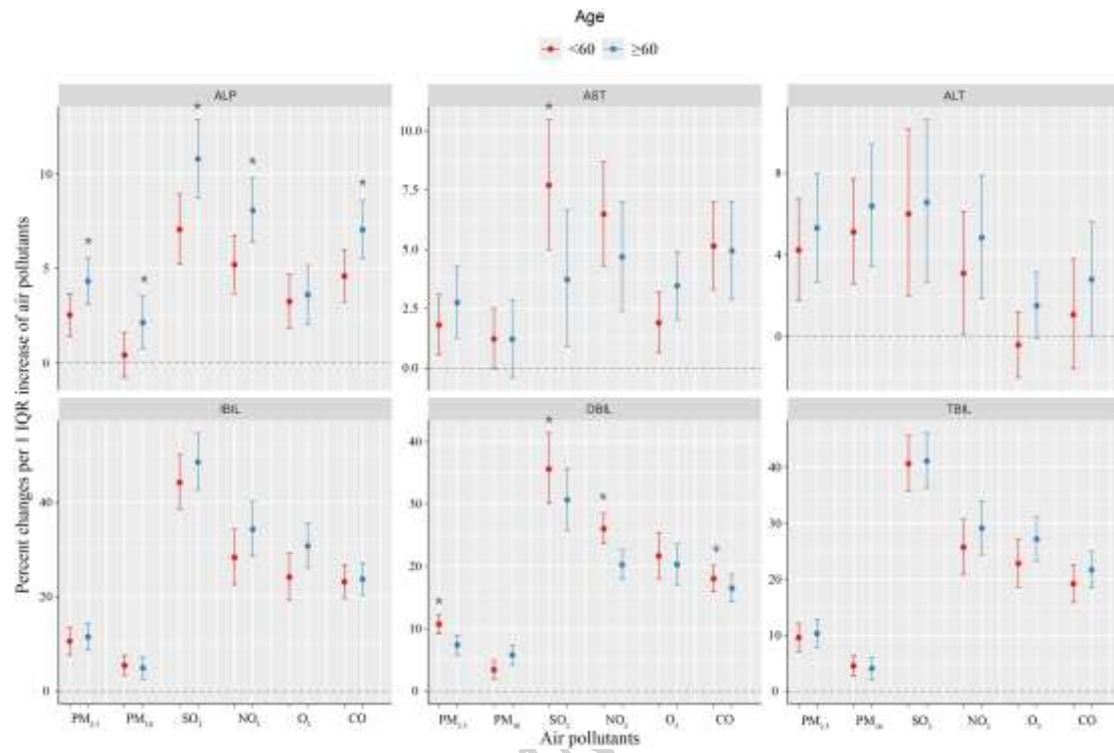
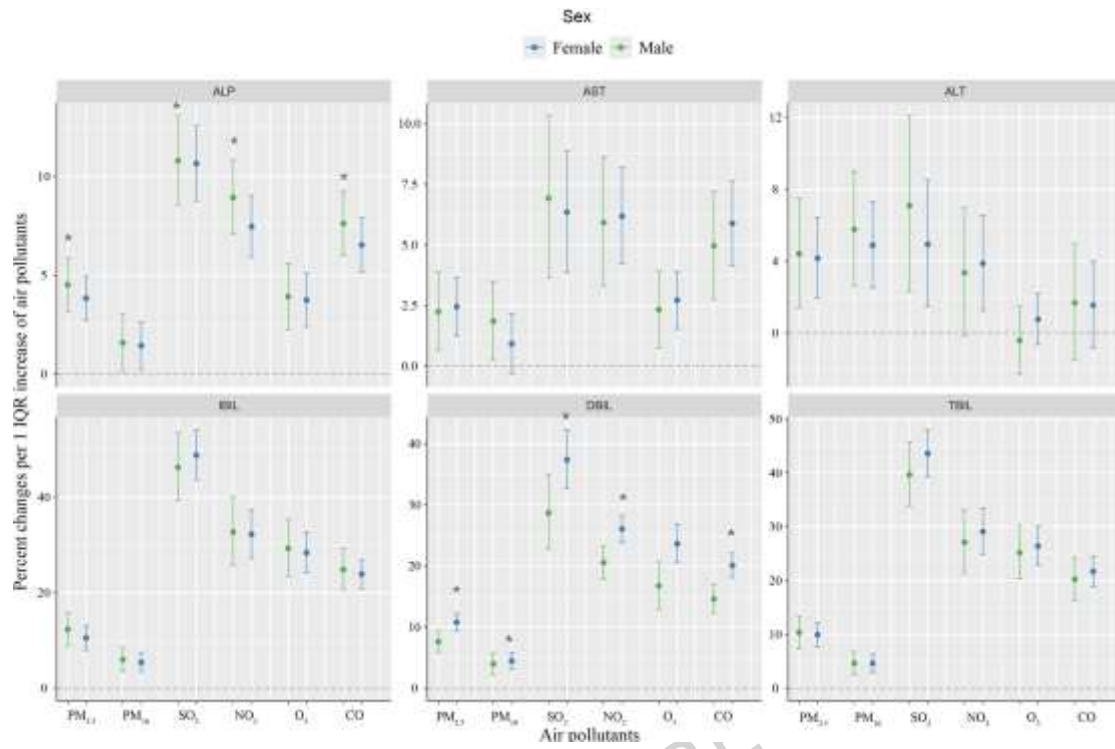
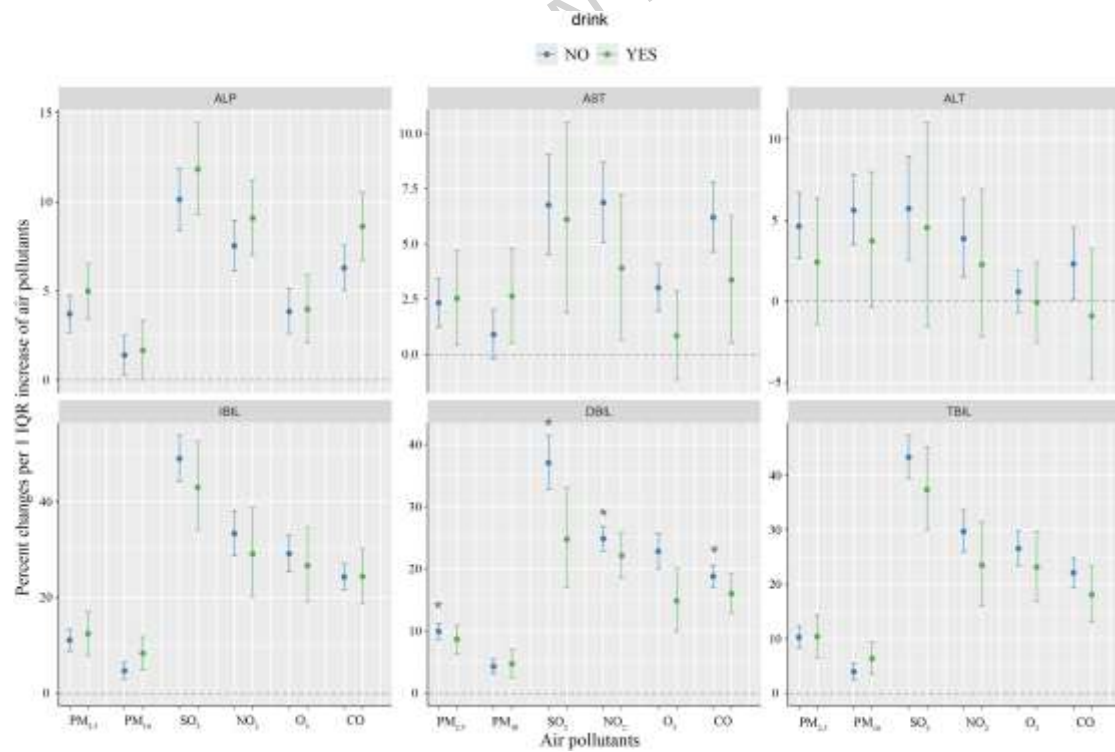
Fig. 3.

Fig. 3 WQS model regression index weights for ALP(A), AST(B), ALT(C), IBIL(D), DBIL(E), and TBIL(F). Models were adjusted for all covariates. PM_{2.5}, particulate matter with an aerodynamic diameter $\leq 2.5 \mu\text{m}$; PM₁₀, particulate matter with an aerodynamic diameter $\leq 10 \mu\text{m}$; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; IBIL, indirect bilirubin; DBIL, direct bilirubin; TBIL, total bilirubin.

Fig. 4

A



B**C**

D

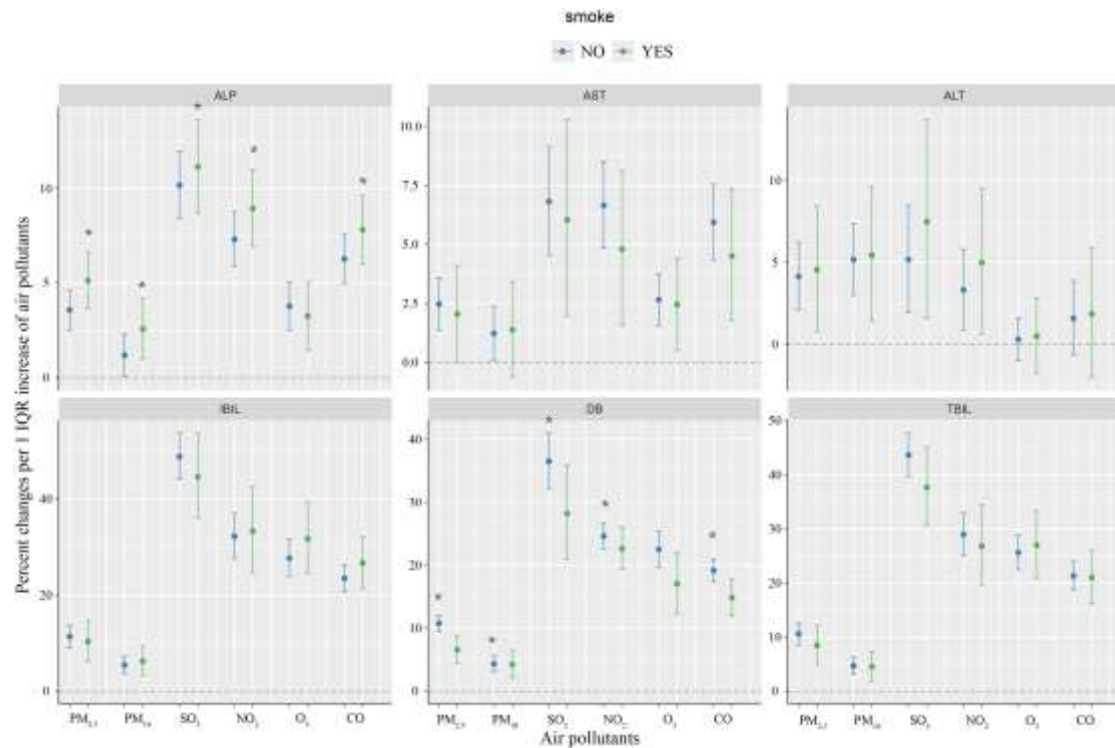


Fig. 4 Stratified analyses of age (A), sex (B), drinking history (C) and smoking history (D) on the association between air pollutants and biomarkers of liver function. PM_{2.5}, particulate matter with an aerodynamic diameter ≤ 2.5 μm ; PM₁₀, particulate matter with an aerodynamic diameter ≤ 10 μm ; SO₂, sulfur dioxide; NO₂, nitrogen dioxide; O₃, ozone; CO, carbon monoxide; AST, aspartate aminotransferase; ALT, alanine aminotransferase; ALP, alkaline phosphatase; IBIL, indirect bilirubin; DBIL, direct bilirubin; TBIL, total bilirubin.

*: It denoted statistical significance after Benjamini-Hochberg correction for multiple comparisons.